

TIM YANG

University of Illinois
Siebel School of Computing and Data Science
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EDUCATION

University of Illinois Urbana-Champaign (UIUC)

PhD, Kinesiology	Computational Rehab for Power Wheelchair Users	≈2025
Certificate	Information Accessibility, Design, and Policy	2019
Certificate	Foundations in Teaching	2017

University of Central Oklahoma

BS, Computer Science	2012
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POSITIONS

Siebel School of Computing and Data Science, UIUC

Senior Instructional Designer	2023+
Instructional Designer	2020–2023

Microsoft Lighthouse Program, UIUC

Doctoral Fellow	2018–2020
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Department of Health and Kinesiology, UIUC

Teaching Assistant	2015–2017
Research Assistant	2012–2017

Department of Rehab Sciences, University of Oklahoma

Research Assistant	2011–2012
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HONORS

Fellowships

Doctoral Fellowship	Microsoft Lighthouse Program, UIUC	2018–2020
Research Scholarship	NIH INBRE Program, University of Oklahoma	Su 2011

Awards

Finalist	Image of Research Competition, UIUC	2019
1 st Place	Research Live Competition, UIUC	2018
Best Paper	Student Paper Competition, RESNA Conference	2015
Honorable Mention	Student Paper Competition, RESNA Conference	:
2 nd Place	Computational Science and Engineering Competition, UIUC	2014
Honorable Mention	Student Paper Competition, RESNA Conference	:
Outstanding Research	Department of Computer Science, UCO	2012

RESEARCH

Power Wheelchair Mobility

Risk model	Predicted wheelchair driving risk via probabilistic AI/ML
Joystick model	Modeled muscle synergies for power wheelchair driving via AI/ML
Cloud monitor	Monitored power mobility usage via custom cloud dashboard
Wheelchair tracker	Tracked free-living wheelchair usage via accelerometry

Power Wheelchair Seating

Risk model	Predicted pressure ulcer risk from seat pressure via AI/ML
Personalized cushion	Assessed inflation patterns via custom programmable cushion
Tilt/recline	Assessed seat pressure vs tilt/recline via ANOVA

Pressure Ulcer Pathophysiology

Skin blood flow dynamics	Assessed reactive hyperemia in spinal injury via laser Doppler
Soft tissue mechanics	Assessed skin blood flow response vs mechanical indentation in spinal injury
Biomarker profiles	Tested cytokine response vs ulceration in rats via ELISA

PUBLICATIONS

Journals

Ren S, Chen Z, Qin X, Zhao X, Yang TD, Zhu W. Measurement and evaluation of bone loading in physical activity: a systematic review. *Meas Phys Educ Exerc Sci*. 2021;25(2):[149–162](#)

Yang TD, Jan YK. Predicting pressure ulcer risk from seating interface pressure using nonnegative matrix factorization. *Med Biol Eng Comput*. 2020;58:[227–237](#)

Lung CW, Yang TD, Liao BY, Cheung WC, Jain S, Jan YK. Dynamic changes in seating pressure gradients in people with spinal cord injury. *Assist Technol*. 2020;32(5):[277–286](#)

Liao F, Yang TD, Wu FL, Cao CM, Mohamed A, Jan YK. Using multiscale entropy to assess the efficacy of local cooling on reactive hyperemia in people with spinal cord injury. *Entropy*. 2019;21(1):[90](#) (12 pages)

Chen Y, Wang J, Lung CW, Yang TD, Crane B, Jan YK. Effect of wheelchair tilt and recline on ischial and coccygeal interface pressure in people with spinal cord injury. *Am J Phys Med Rehabil*. 2015;93(12):[1019–1030](#)

Lung CW, Yang TD, Crane B, Elliott J, Dicianno BE, Jan YK. Investigation of peak pressure index parameters for people with spinal cord injury using wheelchair tilt and recline: methodology and preliminary report. *Biomed Res Int*. 2014;2014:[508583](#) (9 pages)

Yang TD, Hutchinson SA, Rice LA, Watkin KL, Jan YK. Development of a scalable wireless monitoring system for wheelchair tilt usage. *Int J Phys Med Rehabil*. 2013;1(4):[129](#) (6 pages)

Proceedings

Yang TD, Rice LA, Hutchinson SA, Jan YK. Robotic Individualized Driving Evaluation (RIDE): design and preliminary evaluation. *Proceedings of the 19th IEEE International Conference on Rehab Robotics (ICORR)*; May 2025; Chicago, IL (6 pages)

Yang TD, Rice LA, Jan YK. Typifying power wheelchair joystick control using EMG feature engineering and visualization. *Proceedings of the 39th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; July 2018; Arlington, VA (4 pages)

Yang TD, Rice LA, David A, Hutchinson SA, Jan YK. Myoelectric modeling of joystick control for adaptive smart wheelchairs. *Proceedings of the 36th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; June 2015; Denver, CO (4 pages)

- **Award: Best Paper**

Yang TD, Kibler K, Lung CW, Jan YK. Development and evaluation of a programmable alternating pressure seat cushion. *Proceedings of the 36th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; June 2015; Denver, CO (4 pages)

- **Award: Honorable Mention Paper**

Yang TD, Patil A, Jan YK. Individualized performance quantification of power wheelchair driving. *Proceedings of the 35th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; June 2014; Indianapolis, IN (4 pages)

- **Award: Honorable Mention Paper**

PRESENTATIONS

Talks

Yang TD. On the road to self-driving ... drivers? Presented at: *Research Live Competition*; October 2018; Urbana, IL

- **Award: 1st Place**

Yang TD, Rice LA, David A, Hutchinson SA, Jan YK. Myoelectric modeling of joystick control for adaptive smart wheelchairs. Presented at: *36th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; June 2015; Denver, CO

Yang TD, Patil A, Jan YK. Individualized performance quantification of power wheelchair driving. Presented at: *35th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; June 2014; Indianapolis, IN

Yang TD, Liao F, Jones MA, Jan YK. Effect of wheelchair tilt and recline on peak seating pressure in people with spinal cord injury. Presented at: *28th Southern Biomedical Engineering Conference (SBEC)*; May 2012; Houston, TX

Posters

Yang TD, Rice LA, Hutchinson SA, Jan YK. Robotic Individualized Driving Evaluation (RIDE): design and preliminary evaluation. Presented at: *19th IEEE International Conference on Rehab Robotics (ICORR)*; May 2025; Chicago, IL

Yang TD, Rice LA, Jan YK. Typifying power wheelchair joystick control using EMG feature engineering and visualization. Presented at: *39th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; July 2018; Arlington, VA

Jan YK, Lung CW, Yang TD, Cheung W, Jain S. Seating pressure gradient vectors in response to wheelchair tilt and recline in people with spinal cord injury. Presented at: *93rd American Congress on*

Rehab Medicine (ACRM); November 2016; Chicago, IL

Yang TD, Kibler K, Lung CW, Jan YK. Development and evaluation of a programmable alternating pressure seat cushion. Presented at: *36th Rehab Engineering and Assistive Technology Society of North America (RESNA) Conference*; June 2015; Denver, CO

Yang TD, Hutchinson SA, Jan YK. Markov framework for power wheelchair driving. Presented at: *3rd Computational Science and Engineering (CSE) Meeting*; April 2014; Urbana, IL

- **Award: 2nd Place**

Yang TD, Hutchinson SA, Rice LA, Watkin KL, Jan YK. Pressure ulcer prevention via Raspberry Pi. Presented at: *Center on Health, Aging, and Disability Symposium*; March 2013; Champaign, IL

Yang TD, Liao F, Jones MA, Jan YK. Sitting-induced pressure ulcer risks may be reduced at specific tilt and recline angles. Presented at: *NIH IDeA Networks of Biomedical Research Excellence (INBRE) Symposium*; July 2012; Oklahoma City, OK

Yang TD, Liao F, Jones MA, Jan YK. Effect of wheelchair tilt and recline on peak seating pressure in people with spinal cord injury. Presented at: *Allied Health Research Day*; April 2012; Oklahoma City, OK

Yang TD, Liao F, Jones MA, Jan YK. Effect of wheelchair tilt and recline on peak seating pressure in people with spinal cord injury. Presented at: *Graduate Research Education and Technology Symposium*; April 2012; Oklahoma City, OK

Yang TD, Fu J, Jones MA, Jan YK. Quantifying free-living power wheelchair usage using accelerometry. Presented at: *Oklahoma Research Day*; November 2011; Lawton, OK

TEACHING

Center for Innovation in Teaching and Learning, UIUC

Guest Lecturer	Title II Accessibility Workshop Series	2025
Guest Lecturer	Graduate Academy for College Teaching	2020
Guest Lecturer	Teaching Professionals Program (TPR02)	2018

Department of Health and Kinesiology, UIUC

Lab Instructor	Rehab Biomechanics (KIN494)	2015–2017
Teaching Assistant	Drug Use and Abuse (CHLH243)	2015

International Graduate Mentor Program, UIUC

Mentor	6 grad students (Rehab Science)	2018–2019
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Independent Study, UIUC

Mentor	2 undergrad students (Kinesiology)	2018
Mentor	1 undergrad student (Mechanical Engineering)	2014

MoST Scholars Program, UIUC

Mentor	6 undergrad students (Biomedical Engineering)	2014–2016
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Khorana Scholars Program, UIUC

Mentor	1 grad student (Bioengineering)	Su 2014
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GRANTS

Completed

13288 UIUC Campus Research Board. Wheelchair tilt and recline for preventing pressure ulcers in people with spinal cord injury. Cost: \$25,000. PI: Jan YK; 2013–2014

- Role: Research Assistant (first/co-authored 2 peer-reviewed publications)

R03HD060751 NIH Eunice Kennedy Shriver National Institute of Child Health and Human Development. Effect of power seating on tissue viability in wheelchair users with spinal cord injury. Cost: \$165,500. PI: Jan YK; 2011–2012

- Role: Research Assistant (co-authored 2 peer-reviewed publications)

Unfunded

D2 NIDILRR Rehab Engineering Research Center. Development of a pervasive power seating framework. PI: Hutchinson SA, Jan YK; 2013

- Role: Research Assistant (drafted complete proposal)

D1 NIDILRR Rehab Engineering Research Center. Development of a pervasive power mobility framework. PI: Hutchinson SA, Jan YK; 2013

- Role: Research Assistant (drafted complete proposal)

SERVICE

Professional

Member	IEEE (Institute of Electrical and Electronics Engineers)	2024+
Member	EMBS (IEEE Engineering in Medicine and Biology Society)	2024+
Member	ICORR (IEEE International Consortium of Rehab Robotics)	2024+
Member	RESNA (Rehab Engineering Society of North America)	2013–2019

Intramural

Accessibility Liaison	Illinois Accessibility Liaison Program, UIUC	2019+
Judging Committee	Research Live Competition, UIUC	2018
English Coach	Gies College of Business, UIUC	:
Grader	Illinois Math Finals, Department of Mathematics, UIUC	2014–2016

Extramural

Reviewer	ICORR Conference	2024+
Reviewer	PLOS ONE	2018+
Reviewer	RESNA Conference	2015–2019